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VR-ManuSafe: Virtual reality learning environment for manufacturing safety training



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ABSTRACT

This paper presents VR-ManuSafe, a VR-based safety training model for the manufacturing industry developed in Unity, which adopts a modular design and provides highly simulated safety training scenarios. The scenarios cover eight common injury types in the manufacturing industry and are combined with safety learning and online assessment modules to enhance and monitor training effectiveness. A trial with 35 participants showed that VR training significantly improved concentration, confidence and satisfaction compared to traditional methods. This highlights the potential of VR to enhance learning and user experience in safety training.

Code metadata

Current code version	*V12*
Permanent link to code/repository used for this code version	https://github.com/SoftwareImpacts/SIMPAC-2025-123
Permanent link to reproducible capsule	
Legal code license	*MIT License*
Code versioning system used	*none*
Software code languages, tools and services used	*C#, Windows 10/11,Unity 2022.3.49f1*
Compilation requirements, operating environments and dependencies	*Windows 10/11, Microsoft HoloLens 2*
If available, link to developer documentation/manual	https://github.com/EmilyWinnie123/VR-ManuSafe/blob/main/README.md
Support email for questions	p137552@siswa.ukm.edu.my

1. Motivation and significance

Work accidents are unexpected events that cause physical or psychological injury, illness, or death [1]. Sunyoung Park's research indicates that, based on data from 1247 companies in South Korea between 2011 and 2021, the number of non-fatal injuries in the manufacturing sector was 15,915, accounting for 50.17 % of all occupational injuries [2]. In Small and Medium-Sized Enterprises (SMEs), human errors and employee turnover can trigger occupational accidents due to insufficient protective equipment, inadequate occupational safety training, and a lack of safety measures in the workplace [3]. Safety training can help employees better understand risks and safety knowledge, provide safe operating procedures, and significantly reduce "task complexity", thereby minimising the risk of human errors [3,4]. However, traditional safety training methods, such as classroom lectures and on-site demonstrations, although somewhat effective in improving employees' safety awareness, are limited by employees' engagement and practical operation opportunities [5,6]. The emergence of Virtual Reality (VR) technology offers a new solution for safety training. VR technology is capable of creating immersive virtual environments [7], which not only allows learners to access more realistic classroom information [8] but also allows employees to experience a variety of safety risks and emergencies firsthand in virtual scenarios, which significantly improves the effectiveness of training [9,10]. Manufacturing plants face various safety risks during production, such as mechanical

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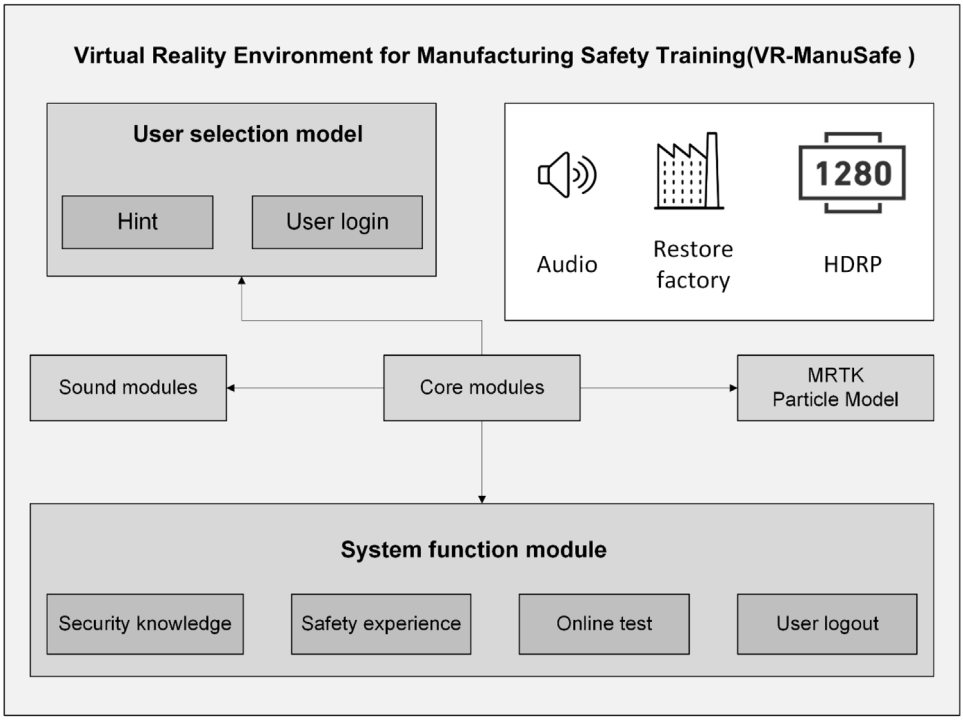


Fig. 1. Core functional framework diagram.

injuries, electrical accidents, and fires. These risks threaten employees' lives and lead to production interruptions and economic losses [11]. Therefore, exploring a more effective and economical safety training method is significant. This paper aims to investigate the effects of VR technology on safety training for small manufacturing plants. The study assesses the system's impact on employees' safety awareness and emergency response capabilities by developing a VR training prototype based on Unity and selecting eight common injury types for training experiences [12].

2. Architecture and functionality of the VR safety training system

As previously introduced in this paper, VR-ManuSafe is a VR-based model that allows users to experience and learn manufacturing safety training concepts in a three-dimensional environment. The system employs high-definition rendering pipeline technology to faithfully reproduce the factory layout, with training content covering common injury types in manufacturing and offering good scalability. First, by replicating real injury scenarios, VR-ManuSafe can effectively convey the importance of safety training and reduce the gap between the virtual training environment and the factory setting. Second, these common injury scenarios apply to various types of factories and provide broad applicability for safety training. Most importantly, the system's scalability allows different factories to customise VR safety training systems based on this universal template [13].

Fig. 1 provides a general overview of the VR-ManuSafe architecture, including the relationships between modules and their main components. The primary modules of this architecture are composed of three parts: Safety Knowledge Module, the Safety Experience Module, and the Online Testing Module. Each module leverages VR technology to immerse trainees in simulation scenarios similar to manufacturing environments, enabling safer and more effective learning and practice of safety knowledge and skills. These modules guide manufacturing safety training through foundational knowledge learning and immersive experiences, organised as a series of structured knowledge modules where the safety experience component requires trainees to interact with various immersive hazards and learn relevant safety knowledge in a virtual factory.

Given the diversity of the manufacturing industry and the need for comprehensive risk coverage, the Safety Experience Module includes eight common injury types prevalent in small, medium, and large manufacturing plants. This approach aims to overcome the limitations of traditional systems that often focus on specific risks, such as fire and gas leak escape scenarios, thereby achieving a more comprehensive risk management solution. Furthermore, through high-precision 3D modelling, lighting effects, and modular design, the system significantly enhances scene realism and system scalability, supporting flexible, functional expansion and diverse application needs. This design facilitates other manufacturing plants in implementing their personalised requirements.

2.1. Function selection module

VR-ManuSafe is based on a VR framework compatible with Unity, designed to run on the Oculus Quest 2 equipped with Android 10. The study selected the Oculus Touch controllers as the interaction tool because they are highly compatible with the Meta Quest series devices, allowing trainees to perform various operations in the virtual environment, such as manipulating equipment and triggering alarms. In addition, as shown in Fig. 2, screenshots of the different interfaces of the VR-ManuSafe system when running on a virtual reality device are presented in full, specifically (a) the operation prompt interface, (b) the function selection interface, (c) the safety knowledge learning interface, (d) the online testing interface, (e, f) the electrocution simulation scenario, and (g, h) the fire simulation scenario.

The Function Selection Module manages the initial menu, which allows users to choose different modules within the system. Subsequently, it directly oversees the system's operations. In this case, the Game Module is responsible for loading levels from the disk and passing them to the Entity Module for processing and rendering. This module lets users select training modules by clicking on the menu options.



Fig. 2. Screenshots of different screens of VR-ManuSafe running on a virtual reality device: (a) tips; (b)function selection; (c) safety knowledge; (d) online test; (e,f) electric shock; (g,h) fire.

2.2. Safety knowledge module

The core function of the Safety Knowledge Module is to provide systematic safety education through virtual reality technology. Trainees can learn about equipment operation, hazard identification, and emergency response procedures in the virtual environment. The module uses click-based navigation to switch between knowledge points, ensuring practical knowledge delivery. Additionally, the system optimises teaching content and methods through real-time user feedback to adapt to industry needs. This module encapsulates all functions related to safety knowledge and allows for creating and expanding personalised safety training content within the system. To this end, the module generates and expands corresponding text and images in JavaScript Object Notation (JSON) format for easy sharing or editing.

2.3. Safety experience module

The Safety Experience Module lets trainees experience safety accident scenarios firsthand in the virtual environment, enhancing their emergency response capabilities. This module simulates a variety of typical workplace safety accidents, such as fire injury and electrical accidents, where trainees can interact, operate, and participate in multi-user collaborative training within these simulated scenarios. Virtual reality technology and a high-definition rendering pipeline ensure that accident simulations are highly realistic, aligning closely with actual working environments.

2.4. Online assessment module

Finally, the Online Assessment Module collects and analyses data in real-time through tests and evaluation tasks to assess trainees' mastery

Table 1
User experience evaluation questions.

No.	Question	Reference
1	*Do you find the system operation intuitive and easy to understand?*	[18]
2	*Is the system interface clear and concise?*	[19]
3	*Can you understand and effectively apply safety rules in virtual accident scenarios?*	[18]
4	*Were you immersed in realistic accident scenarios, and does the VR experience feel authentic?*	[18]
5	*Does the system's interactivity encourage you to participate actively and leave a deep impression of the learning content?*	[19]

of safety knowledge. Assessment results are promptly fed back to trainees and training managers to track learning progress and provide data support for personalised training. The system improves trainees' safety knowledge and emergency handling capabilities through these three modules. It optimises training effectiveness through user feedback, ensuring adaptability and training outcomes in different manufacturing environments.

3. Method

This study conducted a comprehensive usability evaluation of the manufacturing industry's VR safety training system, comparing its performance to traditional training methods. The study aimed to evaluate the practical effectiveness, user engagement, and overall performance of the system by focusing on user experience, ease of use, effectiveness, immersion, and interactivity. The study applied the ARCS model of Attention, Relevance, Confidence, and Satisfaction to evaluate [14] and examined whether VR training improves motivation and learning compared to traditional methods [15]. It also assessed the system's realistic simulation, adaptability and scalability to high-stakes scenarios under different conditions. The study used a five-point Likert scale questionnaire to collect user feedback on the VR system [16]. Based on a previous related study, it was known that 20 users could identify at least 95% of the problems during usability testing [17]. To further enhance the methodological rigour of this study, a cohort of 35 participants was recruited to complete all safety training tasks and provide comprehensive post-training evaluations. The results are intended to identify strengths and weaknesses, guide future improvements, and demonstrate VR's unique value and potential in safety training. As shown in Table 1, a series of questions was designed to evaluate the user experience based on the ARCS model.

4. Results

This study analysed survey questionnaires from 35 participants, ranging from 20 to 52 years old (mean age = 32, standard deviation = 9). In this study sample, males accounted for 60.0% (n = 21) and females for 40.0% (n = 14), with a sex ratio of 1.5:1, indicating a relatively higher proportion of males. Regarding educational background (n = 35), the largest group held a high school degree, comprising 37.1% (n = 13); doctoral degree holders accounted for 25.7% (n = 9); bachelor's degree holders 22.9% (n = 8); master's degree holders 8.6% (n = 3); and junior high school graduates 5.7% (n = 2), showing an overall distribution predominantly concentrated at high school and above. The questionnaire was designed based on the ARCS model (Attention, Confidence, Relevance, and Satisfaction) to evaluate the effectiveness and user experience of VR training compared to traditional training methods. Results show that participants who received VR training scored higher in confidence, satisfaction, attention, relevance, and overall user experience than those who received traditional training.

Table 2
Mean scores and T-test results for traditional vs. VR training.

Dimension	Traditional Training Mean	VR Training Mean	t-value	p-value	Significant
Attention	3.30	4.64	−11.68	1.88×10^{-13}	Yes
Relevance	3.45	4.61	−11.19	6.04×10^{-13}	Yes
Confidence	3.39	4.61	−10.02	1.10×10^{-11}	Yes
Satisfaction	3.3	4.55	−10.72	1.86×10^{-12}	Yes

4.1. Comparison between traditional and VR training

Results are presented in Fig. 3 through box plots, and the respective data analysis is presented visually. A comparative evaluation across four critical dimensions reveals that Virtual Reality training consistently outperforms traditional training methods with statistically superior and more homogeneous outcomes. In the dimension of attention, Virtual Reality training exhibits a narrow and elevated score distribution ranging from 3.0 to 5.0 with a median of 4.6. In contrast, traditional training shows a broader dispersion from 1.0 to 5.0 with a median of 3.4, including a low-end outlier indicative of substantial participant divergence. Similarly, in relevance, Virtual Reality achieves scores between 3.0 and 5.0 and a median of 4.6, surpassing traditional training, which ranges from 2.0 to 5.0 with a median of 3.5. This pattern persists in confidence, where Virtual Reality scores range from 3.0 to 5.0 with a median of 4.6, contrasting with the traditional approach spanning 2.0 to 5.0 and a median of 3.3. Finally, in satisfaction, Virtual Reality records scores from 3.0 to 5.0 and a median of 4.5, exceeding the traditional range of 2.0 to 5.0 with a median of 3.2. The absence of outliers in Virtual Reality results across all categories underscores its exceptional reliability and uniformly positive user experience, affirming its pronounced advantages in training efficacy and learner engagement.

To further validate the differences observed in the box plots, independent samples t-tests were conducted to compare traditional training and VR training across the four core dimensions. The data are shown in Table 2, the results indicated that VR training achieved significantly higher mean scores than traditional training in all dimensions (Attention: $t = -11.68$, $p < .001$; Relevance: $t = -11.19$, $p < .001$; Confidence: $t = -10.02$, $p < .001$; Satisfaction: $t = -10.72$, $p < .001$). All p-values were far below the significance threshold of 0.05, confirming that the observed differences are statistically significant. This statistical evidence confirms the box plot analysis, showing that VR training's superior performance in both median and distribution is robust, highlighting its advantages in training effectiveness and user experience.

4.2. User experience

The study results, as shown in Fig. 4, indicate that the average scores across all dimensions ranged from 4.09 to 4.71, reflecting a generally positive user satisfaction. The highest score was achieved in the system's interactivity, with an average of 4.71, demonstrating strong user engagement and appreciation for the learning content. The system interface clarity and simplicity also scored relatively high at 4.57, indicating users found the design easy to navigate. The system

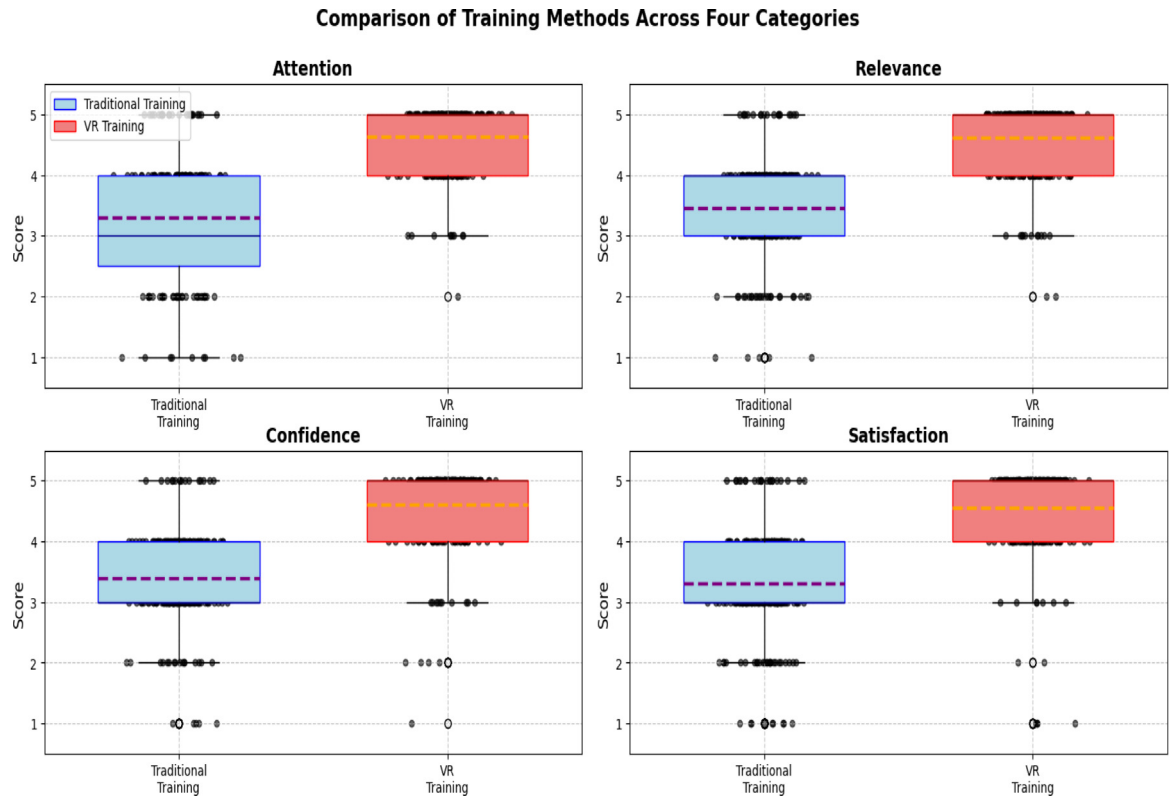


Fig. 3. Comparative analysis of the effect of virtual reality and traditional training based on the ARCS model.

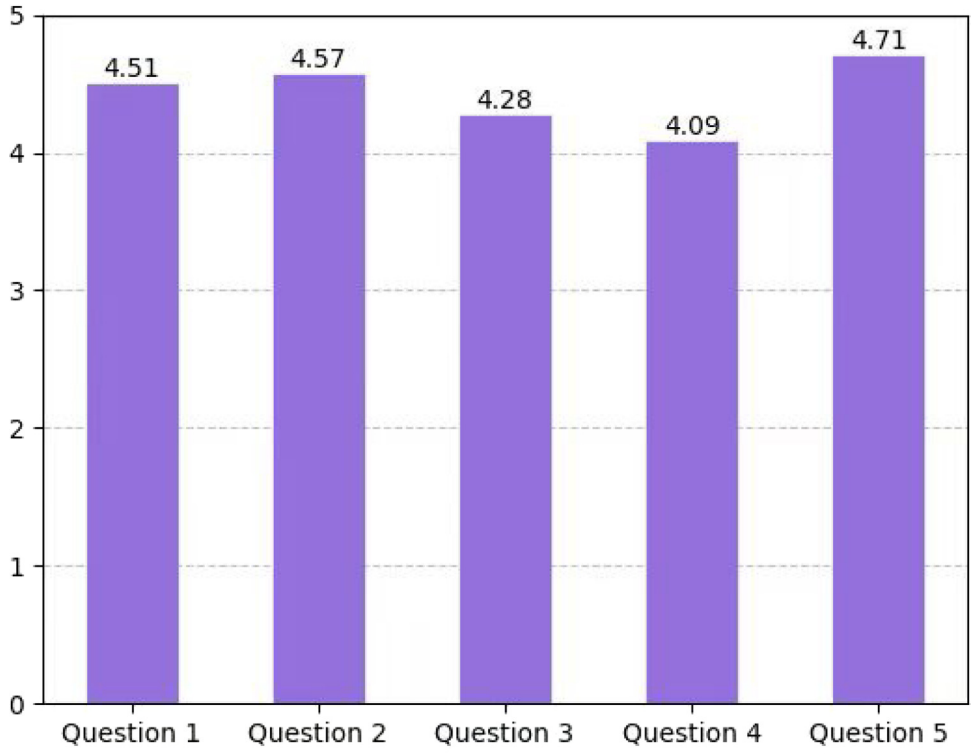


Fig. 4. Statistical chart of average user experience scores for manufacturing safety training systems.

attained a usability score of 4.51, indicating a highly intuitive user experience and efficient operational fluency. However, the dimension assessing the understanding and application of safety rules within the virtual environment had a lower score of 4.28, suggesting that some

users faced difficulties comprehending and applying these rules effectively. The immersive experience's realism scored 4.09, highlighting that while the VR scenarios were somewhat realistic, they could be enhanced further.

5. Software impact

The VR training system proposed in this study demonstrates significant advantages in manufacturing safety, sustainability, cost-effectiveness, and educational innovation. By leveraging a networked model, the system overcomes geographical and temporal constraints [20], providing employees with standardised, high-quality safety training across different locations. VR technology significantly addresses the shortcomings of traditional training methods [21], reducing accident frequencies and optimising training processes, thereby saving related costs. Previous research on applying metaverse technology in manufacturing safety training has predominantly focused on single-scenario simulations, such as fire evacuation and chemical plant leakage escapes, and high-risk industries like mining, where virtual construction of accident environments is relatively straightforward [22,23]. This study seeks to bridge this research gap by shifting the focus to small-scale indoor manufacturing facilities, which involve human–robot interactions and contain distinct safety zones and working areas. By systematically simulating eight common types of workplace injuries, the study establishes a comprehensive safety training framework that can serve as a versatile and cross-industry reference for enhancing hazard identification and response capabilities in small manufacturing settings. The system's immersive experience and real-time feedback mechanism effectively enhance employees' safety awareness and emergency decision-making capabilities, strengthening workplace safety culture. Moreover, applying this system in vocational education offers innovative tools for technical schools and skill development institutions, allowing students to gain practical experience through realistic simulations, enhancing their employability and facilitating technological innovation and digital transformation in the manufacturing sector.

6. Limitations

While the VR safety training system introduced in this study shows significant effectiveness and potential in the manufacturing sector, several limitations are identified. First, although enhancing immersion, the first-person perspective caused mild dizziness in some users, indicating the need for greater flexibility in the system design, such as incorporating alternative viewpoints. Second, the current system primarily focuses on individual training and lacks a dedicated module for team-based training, limiting its applicability in large-scale emergency scenarios that require coordinated efforts. Additionally, the system's scalability is constrained by the high cost of VR hardware. Furthermore, while the system effectively simulates general manufacturing hazards, it falls short in covering niche and industry-specific scenarios.

7. Conclusion and future work

In this paper, the study explored the application of VR technology in manufacturing safety training. The study proposed an immersive, interactive VR training system to enhance employees' safety awareness and emergency response capabilities through highly realistic virtual scenarios. Simulating hazardous work environments allows employees to practice safety procedures risk-free, improving their ability to handle real-world emergencies. The study analysed the system's performance regarding user satisfaction, confidence, and overall experience through feasibility tests. All test results indicate that the VR training system significantly enhances users' safety knowledge, operational confidence, and overall training experience, offering an efficient and innovative solution for manufacturing safety training.

The study hopes to enhance the VR safety training system's flexibility, practicality, and scalability in future work. Specifically, the study could introduce multiple viewpoints to alleviate user discomfort, develop team-based training modules to address complex scenarios, and implement machine learning to deliver personalised training content. Additionally, the content library could be expanded to cover more industry-specific scenarios. Future studies could also explore cloud-based solutions and more affordable hardware options to reduce deployment costs, thereby broadening the system's applicability. These improvements will make the system more efficient, adaptable, and accessible, providing a more innovative and sustainable solution for manufacturing safety training.

CRediT authorship contribution statement

Lei Fengjin: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Lam Meng Chun:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition. **Dalbir Singh:** Conceptualization, Methodology, Validation, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] R. Davies, P. Jones, I. Nuñez, The impact of the business cycle on occupational injuries in the UK, Soc. Sci. Med. 69 (2) (2009) 178–182, http://dx.doi.org/10.1016/j.socscimed.2009.04.033.

[2] S. Park, M.J. Kim, Corporate characteristics and occupational injuries by industry, Saf. Heal. Work. 14 (3) (2023) 259–266, http://dx.doi.org/10.1016/j.shaw.2023.08.004.

[3] C.M. La Fata, L. Adelfio, R. Micale, G. La Scalia, Human error contribution to accidents in the manufacturing sector: A structured approach to evaluate the interdependence among performance shaping factors, Saf. Sci. 161 (2023) 106067, http://dx.doi.org/10.1016/j.ssci.2023.106067.

[4] R. Sharma, D.K. Mishra, The role of safety training in original equipment manufacturing companies on employee perception of knowledge, behavior towards safety and safe work environment, Int. J. Saf. Secur. Eng. 10 (5) (2020) 689–698, http://dx.doi.org/10.18280/ijsse.100514.

[5] F. Caffaro, G. Bagagiolo, M. Cremasco, L. Vigoroso, E. Cavallo, Tailoring safety training material to migrant farmworkers: An ergonomic user-centred approach, Int. J. Env. Res. Public. Heal. 17 (2020) 2104, http://dx.doi.org/10.3390/ijerph17062104.

[6] T. Casey, N. Turner, X. Hu, K. Bancroft, Making safety training stickier: A richer model of safety training engagement and transfer, J. Saf. Res. (2021) http://dx.doi.org/10.1016/j.jsr.2021.06.004.

[7] M.C. Lam, et al., A comparative evaluation of augmented reality and virtual reality on chemistry experiment learning in higher level education, Multimed. Tools Appl. (2024) http://dx.doi.org/10.1007/s11042-024-20490-7.

[8] J. Liu, M.C. Ang, J.K. Chaw, A.-L. Kor, K.W. Ng, M.C. Lam, Assessing the impact and development of immersive VR technology in education: Insights from telepresence, emotion, and cognition, Technol. Forecast. Soc. Change 213 (2025) 124024, http://dx.doi.org/10.1016/j.techfore.2025.124024.

[9] V. Holuša, M. Vaněk, F. Beneš, J. Švub, P. Staša, Virtual reality as a tool for sustainable training and education of employees in industrial enterprises, Sustainability 15 (17) (2023) 2886, http://dx.doi.org/10.3390/su151712886.

[10] A. Gupta, K. Varghese, Scenario-based construction safety training platform using virtual reality, in: Proceedings of the 37th International Symposium on Automation and Robotics in Construction, ISARC, International Association for Automation and Robotics in Construction (IAARC), 2020.

[11] L. Putranti, A.Y. Saputri, P.N. Aryandha, H.D. Herdiany, Manufacturing companies' risk perception based on the influence of safety leadership, safety knowledge, and safety attitudes, J. Bus. Inf. Syst. 5 (2) (2023) http://dx.doi.org/10.36067/jbis.v5i2.219.

[12] L. Fengjin, D. Singh, Virtual reality safety training framework in manufacturing industry, in: Faculty of Information Science and Technology (FTSM), Universiti Kebangsaan Malaysia, UKM, Bangi, Selangor, Malaysia, 2025, PS-FTSM-2025-P137552.

[13] A. Damianou, et al., Towards graph foundation models for personalization, in: Companion Proceedings of the ACM Web Conference 2024, Assocation for Computing Machinery, New York, NY, USA, 2024, pp. 1798–1802, http://dx.doi.org/10.1145/3589335.3651980.

[14] K. Li, J.M. Keller, Use of the ARCS model in education: A literature review, Comput. Educ. 122 (2018) 54–62, http://dx.doi.org/10.1016/j.compedu.2018.03.019.

[15] B. Jongbloed, N. Chaker, Immersive procedural training in virtual reality: A systematic literature review, Comput. Educ. 196 (2024) 105124, http://dx.doi.org/10.1016/j.compedu.2024.105124.

[16] I.E. Allen, C.A. Seaman, Likert scales and data analyses, Qual. Prog. 40 (7) (2007) 64–65.

[17] L. Faulkner, Beyond the five-user assumption: Benefits of increased sample sizes in usability testing, Behav. Res. Methods Instrum. Comput. 35 (2003) 379–383.

[18] H. Kang, J. Yang, B.-S. Ko, B.-S. Kim, O.-Y. Song, S.-M. Choi, Integrated augmented and virtual reality technologies for realistic fire drill training, IEEE Comput. Graph. Appl. 44 (2) (2024) 89–99, http://dx.doi.org/10.1109/MCG.2023.3303028.

[19] J. Dong, D. Li, K. Ozcan, D. Wan, W. Jiang, Y. Chen, Development of cryovr, a virtual reality training system for hands-on cryoem operations, Acta Crystallogr. Sect. Struct. Biol. 78 (7) (2022) 903–910, http://dx.doi.org/10.1107/S2059798322005654.

[20] M.K. Ario, et al., Towards an implementation of immersive experience application for marketing and promotion through virtual exhibition, Softw. Impacts 14 (2022) 100439, http://dx.doi.org/10.1016/j.simpa.2022.100439.

[21] D. King, S. Tee, L. Falconer, C. Angell, D. Holley, A. Mills, Virtual health education: Scaling practice to transform student learning: Using virtual reality learning environments in healthcare education to bridge the theory/practice gap and improve patient safety, Nurse Educ. Today 71 (2018) 7–9, http://dx.doi.org/10.1016/j.nedt.2018.07.009.

[22] R. Toyoda, F. Russo-Abegão, J. Glassey, Vr-based health and safety training in various high-risk engineering industries: a literature review, Int. J. Educ. Technol. High. Educ. 19 (1) (2022) 42, http://dx.doi.org/10.1186/s41239-022-00349-3.

[23] T. Kaarlela, S. Pieska, T. Pitkaaho, Digital twin and virtual reality for safety training, in: 2020 11th IEEE International Conference on Cognitive Infocommunications (CogInfoCom), IEEE, Mariehamn, Finland, 2020, pp. 000115–000120, http://dx.doi.org/10.1109/CogInfoCom50765.2020.9237812.